

Supporting Information for:
Bayesian Multilevel Modeling of Retention Data
Informed by Structural Similarity of Analytes

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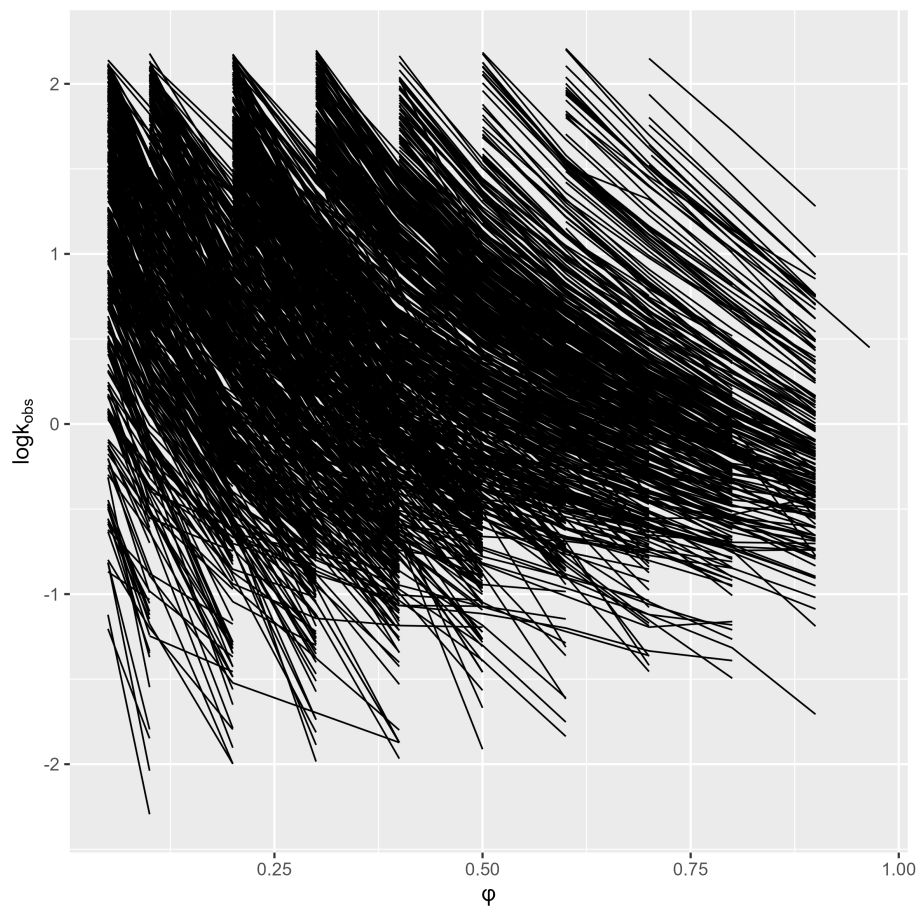
1 Table S1. Summary of the MCMC simulations of the marginal posterior distributions of population-level parameters

variable	mean	median	sd	mad	q5	q95	rhat
θ_{logkw}	3.05	3.05	0.19	0.19	2.74	3.36	1.00
θ_{S1}	3.67	3.67	0.16	0.16	3.40	3.94	1.00
θ_{dlogkw}	-0.35	-0.35	0.12	0.12	-0.55	-0.15	1.00
θ_{dS1}	0.30	0.30	0.09	0.09	0.16	0.45	1.00
β_{logkw}	0.86	0.86	0.05	0.06	0.77	0.95	1.00
β_{S1}	0.62	0.62	0.05	0.05	0.54	0.70	1.00
θ_{S2}	2.90	2.90	0.03	0.03	2.85	2.95	1.02
ω_{logkw}	1.41	1.41	0.04	0.04	1.35	1.47	1.00
ω_{S1}	1.13	1.13	0.03	0.03	1.08	1.18	1.00
$\rho_{1,2}$	0.88	0.88	0.01	0.01	0.87	0.90	1.00
σ	0.05	0.05	0.00	0.00	0.05	0.05	1.00
α	0.71	0.71	0.04	0.04	0.64	0.77	1.00
$\kappa1_{logkw}$	0.15	0.14	0.02	0.02	0.11	0.18	1.01
$\kappa1_{S1}$	0.13	0.13	0.02	0.02	0.10	0.17	1.01
$\kappa2_{logkw}$	0.34	0.33	0.06	0.06	0.23	0.44	1.00
$\kappa2_{S1}$	0.19	0.18	0.07	0.07	0.08	0.30	1.01

mean denotes the posterior mean; median denotes the posterior median, sd denotes the posterior standard deviation; mad denotes the median absolute deviation; q5 denotes the 5th percentile of the posterior draws; q95 denotes the 95th percentile of the posterior draws; rhat denotes the Gelman–Rubin convergence diagnostic. Values close to 1.0 indicate that the MCMC chains have mixed and converged well.

2 Figure S1. Raw data

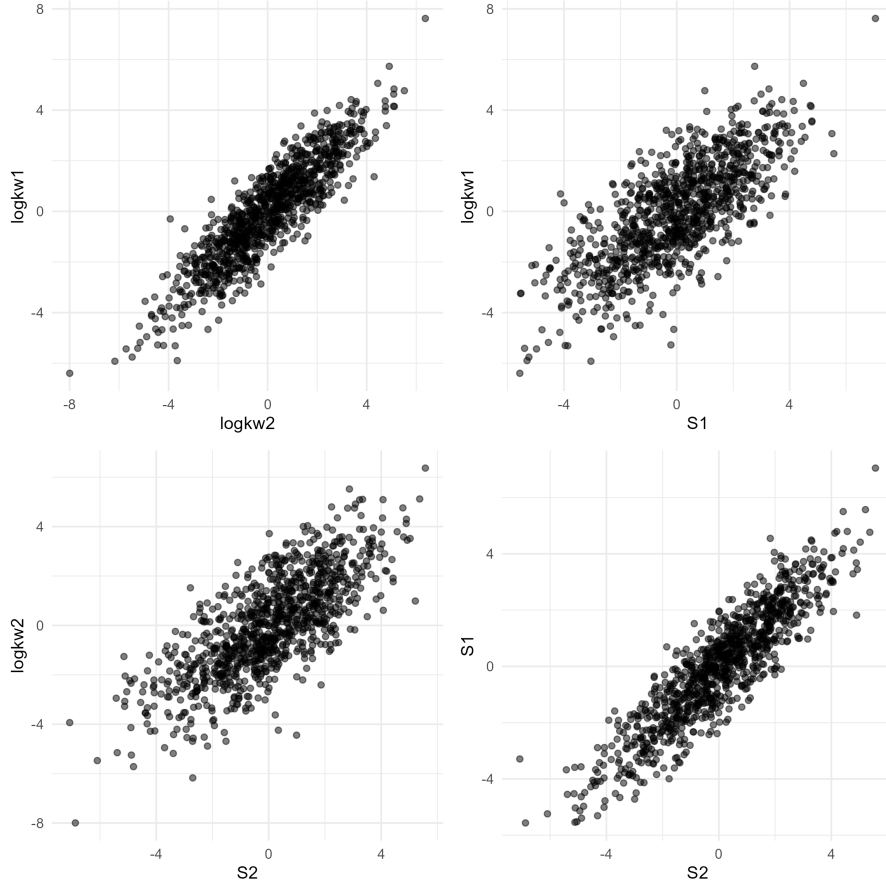
The relationship between observed logk and organic modifier content for all the analytes.



3 Figure S2. Matrix Normal Distribution

Simulations from the Matrix Normal Distribution with row correlations of 0.9 and column correlation of 0.75.

$$\begin{bmatrix} \log k_{w1} & S_1 \\ \log k_{w2} & S_2 \end{bmatrix} \sim \mathcal{MN} \left(\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0.75 \\ 0.75 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right)$$



4 Figure S3. Functional group effects

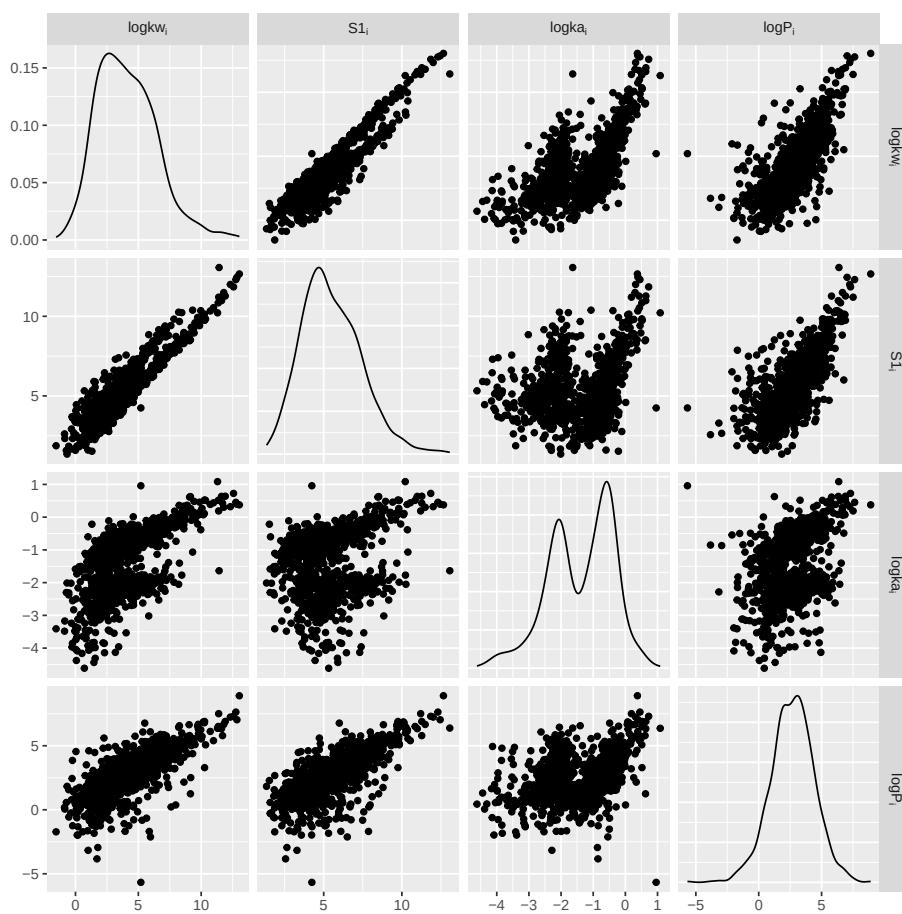
A summary of the marginal posterior distributions of the functional group effects. Red represents the effect of dissociated functional groups, and green represents the effect of non-dissociated functional groups. The least common functional groups (≤ 10 across all analytes) are presented. The plot for the remaining functional groups is presented in Figure 2.



5 Figure S4. Individual parameters, R_i

$\log k_a$ is the (extrapolated) log retention in 100% ACN:

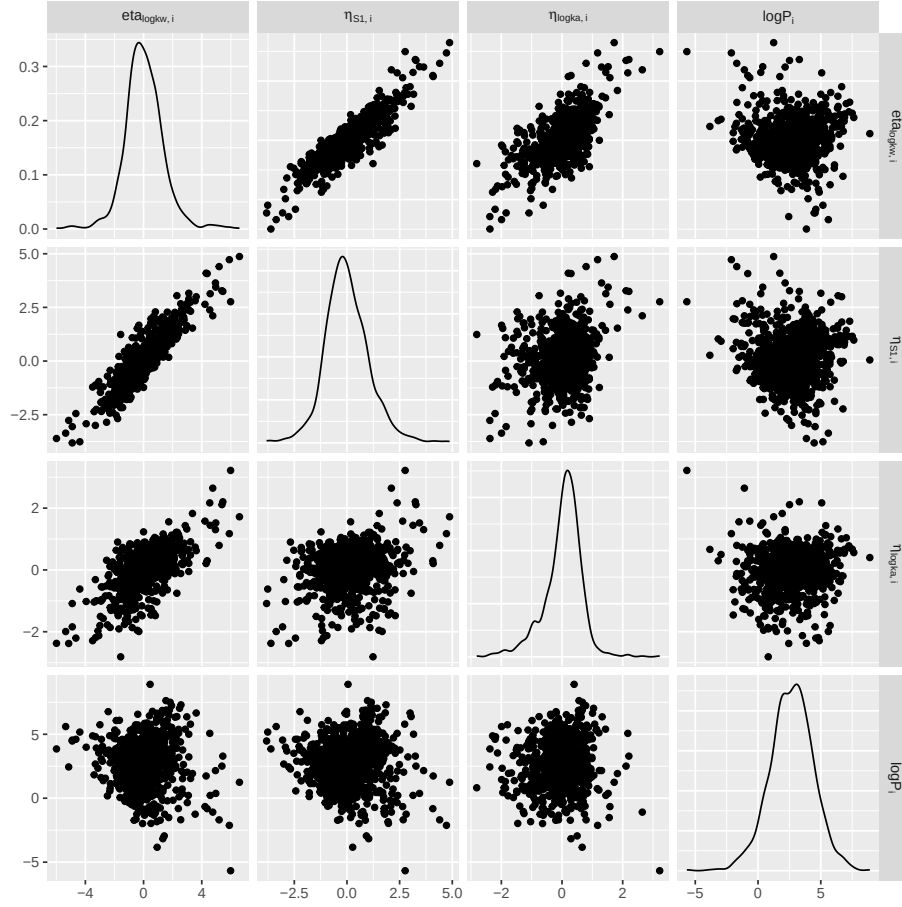
$$\log k_{a_i} = \log k_{w_i} - S_{1,i}$$



6 Figure S5. Individual parameters, $\text{eta}_{R,i}$

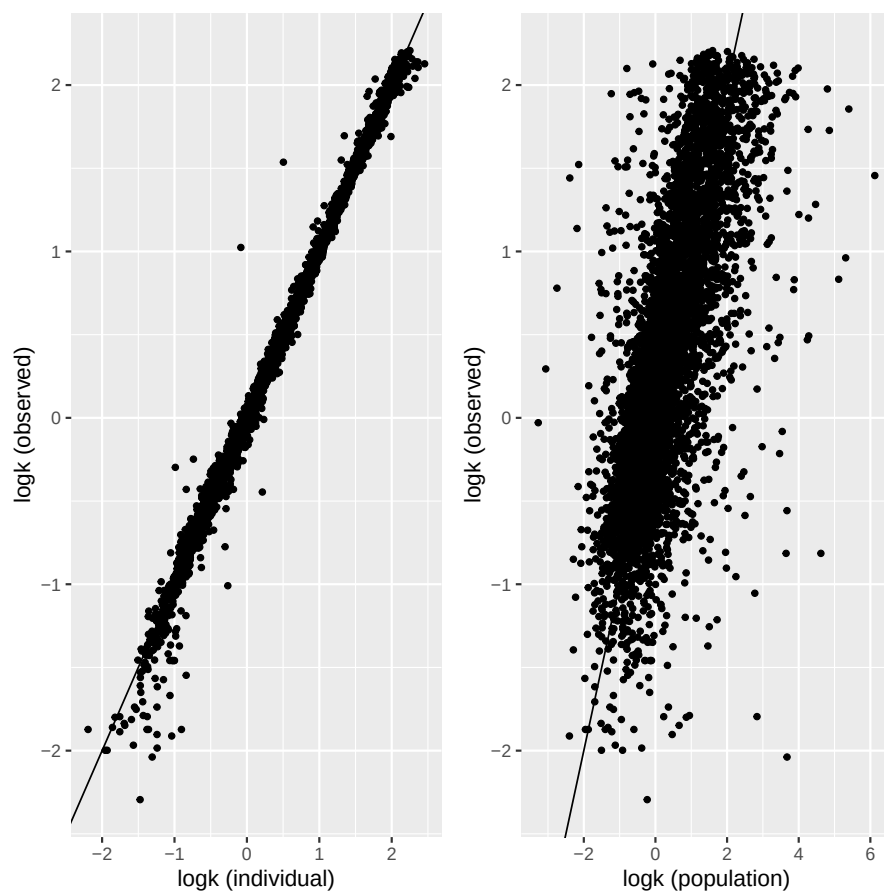
$\text{eta}_{R,i}$ represents the deviation of an individual analyte's parameter from its expected (population-level) value.

$$\eta_{R,i} = R_i - (\theta_R + \beta_R \cdot \log P_i + \pi_{R1} \cdot X_{1i} + \pi_{R2} \cdot X_{2i})$$



7 Figure S6. Goodness of fit plots (GOF)

Observed vs. population and individual predictions:



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Original Computing Environment

R version 4.3.1 (2023-06-16 ucrt)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19045)

Matrix products: default

locale:

```
[1] LC_COLLATE=Polish_Poland.utf8 LC_CTYPE=Polish_Poland.utf8
[3] LC_MONETARY=Polish_Poland.utf8 LC_NUMERIC=C
[5] LC_TIME=Polish_Poland.utf8
```

time zone: Europe/Warsaw

tzcode source: internal

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] cmdstanr_0.8.0.9000 kableExtra_1.4.0    dplyr_1.1.4
```

loaded via a namespace (and not attached):

```
[1] jsonlite_1.8.8      compiler_4.3.1      tidyselect_1.2.1
[4] xml2_1.3.6          stringr_1.5.1       systemfonts_1.1.0
[7] scales_1.3.0        textshaping_0.4.0   yaml_2.3.8
[10] fastmap_1.2.0       R6_2.5.1            generics_0.1.3
[13] distributional_0.4.0 knitr_1.46          backports_1.5.0
[16] checkmate_2.3.1     tibble_3.2.1        munsell_0.5.1
[19] svglite_2.1.3       pillar_1.9.0        posterior_1.6.0
[22] rlang_1.1.3         utf8_1.2.4          stringi_1.8.4
[25] xfun_0.44           viridisLite_0.4.2   cli_3.6.2
[28] withr_3.0.2         magrittr_2.0.3      ps_1.7.6
[31] processx_3.8.4      digest_0.6.35       rstudioapi_0.16.0
[34] lifecycle_1.0.4     vctrs_0.6.5         data.table_1.15.4
[37] tensorA_0.36.2.1    evaluate_1.0.3      glue_1.7.0
[40] codetools_0.2-19    ragg_1.3.2          abind_1.4-8
[43] fansi_1.0.6         colorspace_2.1-0    rmarkdown_2.27
[46] matrixStats_1.3.0   tools_4.3.1         pkgconfig_2.0.3
```

[49] `htmltools_0.5.8.1`